EARTH AND PLANETARY SCIENCE LETTERS 15 (1972) 145–156. NORTH-HOLLAND PUBLISHING COMPANY

PALAEOMAGNETIC RESULTS FROM THE PAGANZO BASIN OF NORTH-WEST ARGENTINA

R. THOMPSON

Department of Geophysics and Planetary Physics, The University, Newcastle upon Tyne, NE1 7 RU, UK

Received 31 January 1972
Revised version received 11 March 1972

The remanent magnetization of red sandstone and siltstone samples of Middle/Upper Carboniferous to Triassic age from four localities in the Paganzo Basin has been measured. Field and laboratory tests demonstrate the stability of the remanence of the thermally cleaned six formation results. The results span the Kiaman Reversed Magnetic Interval. The directions of magnetization confirm previous results from South America, but indicate that the available palaeomagnetic results from the southern continents do not confirm, in detail, existing Gondwana reconstructions.

1. Introduction

The Paganzo Basin lies on the continental side of the Andes in north-west Argentina (fig. 1). Palaeomagnetic studies in the area were first carried out by Creer in 1958 [1]. Following this work more detailed collections, particularly of Middle Paganzo sediments, were made in 1967 by Embleton [2, 3]. These rocks were subjected to thermal demagnetization and were shown to be highly suitable for palaeomagnetic investigation. In 1969 and 1970, further collections were made at new localities in order to give a more complete palaeomagnetic record for the interval of time covered by Paganzo sedimentation.

K-Ar and palaeomagnetic results from an investigation of a basalt in the sequence have been presented previously, Thompson and Mitchell [4]. The palaeomagnetic results from the sandstone collections are presented below and compared with results from other continents.

2. Geology

Deposition of sediments in the Paganzo Basin began after the Acadian-Breton phase of the Carboniferous Geosyncline to the west of the Basin [5]. During the early period of deposition material was provided by local sources, and depressions in the Basin were almost completely filled with conglomerates and coarse sandstones. Marine intercalations are found in the Lower Paganzo sequences, but correlation is still difficult between local centres. Middle Paganzo sediments are similar throughout the Basin, consisting typically of cross-bedded siltstones and fine-grained red sandstones, commonly with a calcareous cement; the total thickness of sediments is around 1500 m. Basaltic flows and ash bands are found at several horizons. Sediments of the Upper section are again more varied, due to renewed tectonic movements in both the source areas and in the Basin itself. The thickness and type of these terrestrial deposits differ from area to area and their age is uncertain, although generally referred to as Triassic. Following detailed mapping of the central part of the Paganzo Basin, Azcuy and Morelli have divided the Paganzo sequence into three formations: Lagares, La Colina and Amana [5] as shown in table 1. The Paganzo sequences were later faulted and folded along reactivated lines of weakness in the basement rocks. These movements were probably associated with the Tertiary orogenesis of the Andean mountain chain.

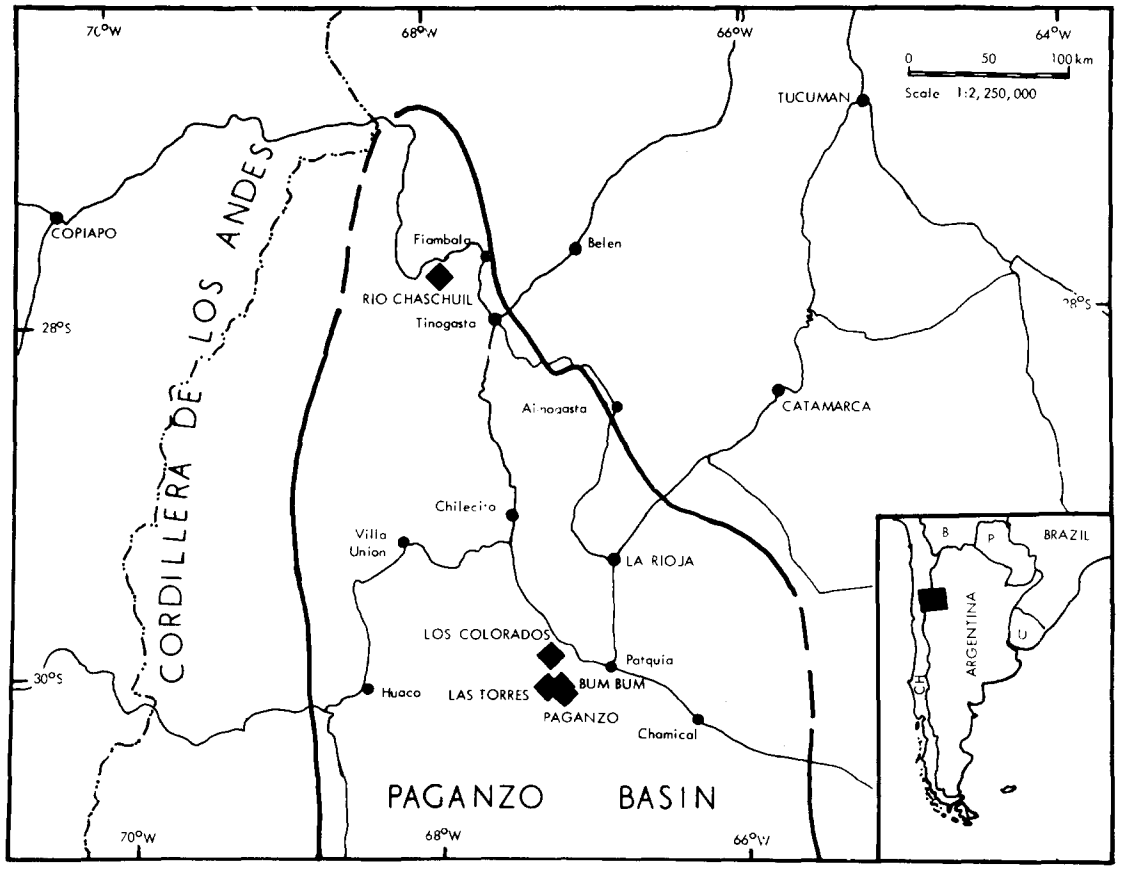


Fig. 1. Collecting sites in the Paganzo Basin.

Table 1
Stratigraphic relationships in the Paganzo Basin.

Age			Formation
Cenozoic			Quaternary and Tertiary
Permo-Triassic	Paganzo succession	Upper Paganzo	AMANA
Permian		Middle Paganzo	LA COLINA
Carboniferous		Lower Paganzo	LAGARES
Lower Palaeozoic and Pre-Cambrian			Crystalline Basement

3. Collection

During 1969 and 1970 red sediments from the Lagares, La Colina and Amana formations were collected near Paganzo village, close to the type locality of each of the formations. Red sandstones of Lagares age from Bum-Bum, of Middle Paganzo age at Los Colorados and from the north of the Basin in the valley of the Rio Chasquil were also sampled for palaeomagnetic investigations. In addition a basic lava was collected from a stratigraphic horizon a few metres above the base of the La Colina formation, both near Paganzo village and Las Torres (fig. 1). Hand samples were collected at all localities and were, whenever possible, orientated using both sun and magnetic compasses in conjunction with a clinometer.

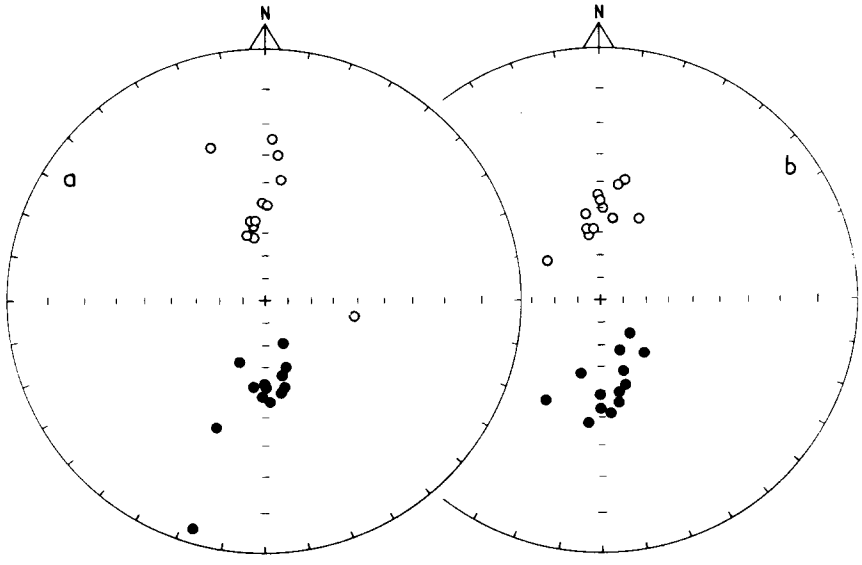


Fig. 2. Bum-Bum. Sample mean directions referred to the palaeo-horizontal; Open circles, negative (up); closed circles, positive (down); (a), natural remanent magnetization; (b), after thermal demagnetization.

4. Laboratory techniques

Whenever possible three 2.54 cm diameter cores were drilled, in the laboratory, from each of the hand samples. One centimetre thick discs were then cut from the cores for measurement of the direction and intensity of the magnetic remanence using the No. 1 sensitive astatic magnetometer in the Nuffield palaeomagnetic laboratory at Newcastle upon Tyne [6]. The initial susceptibility of all specimens was measured using a susceptibility bridge [7].

In order to remove secondary components of magnetization and to test the stability of the remanence, all the sediments were thermally demagnetized. Pilot specimens from each sedimentary sequence were heated to successively higher temperatures and cooled in field-free space; their remanence and susceptibility being measured after each stage. The remaining specimens were demagnetized at a temperature which was determined on the basis of the change of intensity, susceptibility, direction and within-site scatter of the remanence of the pilot specimens. Partial hysteresis ($J_i - H$) and thermal decay of induced magnetization ($J_i - T$) measurements have been made on a magnetic torque balance to help determine the carrier of the magnetic remanence.

In the calculation of the final pole positions all results have been corrected for the dip of the beds from which the samples were collected and are thus referred to the palaeohorizontal. All mean results and their associated errors were derived using the sample as a unit.

5. Bum-Bum

The sediments from this locality are of Lower Paganzo age and belong to the Lagares formation (table 1). They are the first Lower Paganzo rocks to have been measured for palaeomagnetic purposes. Eighteen hand samples were collected from which fiftythree one centimetre thick discs were cut. Both normal and reverse directions were observed in the sediments, the normal magnetizations falling in at least four separate horizons. Seven of the hand samples contain both normal and reverse magnetic directions and a more detailed investigation of these seven samples has been started using a magnetometer designed by Carlinson and de Sa [8] for small scale magnetic studies. The normal and reverse directions have been combined to give a south palaeopole giving unit weight to a normal or reverse group from each hand sample (table 2).

Table 2
Summary of palaeomagnetic results from the Paganzo succession.

Locality	Formation	Rock type	Remanence	Present horizontal					Palaeohorizontal				Polarity	South Palaeopole			
				$N(n)$	D°	I°	csd	α_{95}	D°	I°	csd	α_{95}		Lat.	Long.	csd	α_{95}
Lower Paganzo																	
Bum-Bum	Lagares	Coarse grained red sandstones	NRM	25(53)	177	+55	21.2	7.8	181	+50	20.1	7.4	M	88S	282E	21.5	7.9
Bum-Bum	Lagares		250°C	25(53)	171	+58	17.1	6.2	176	+53	16.7	6.1	M	84S	333E	21.3	7.8
Bum-Bum	Lagares		450°C	25(53)	172	+57	14.5	5.2	177	+52	14.5	5.3	M	85S	335E	18.1	6.6[e] A
Middle Paganzo																	
Las Torres Site A[a]	La Colina	Tholeiitic basalt	300 0e	5(14)	156	+70	6.7	6.4	156	+62	6.7	6.4	R	67S	341E	10.0	9.5
Las Torres Site B[a]	La Colina	Tholeiitic basalt	300 0e	4(12)	184	+63	6.4	7.3	184	+63	6.4	7.3	R	76S	282E	8.0	9.1
Mean of La Colina basalt[b]	La Colina	Tholeiitic basalt	A.F. cleaned	17(50)	152	+75	16.8	7.5	155	+60	11.2	5.0	M	66S	348E	15.5	6.9[e] B
Paganzo Village	La Colina	Fine grained red sandstones	NRM	35(136)	304	+68	66.7	29.8	164	+70	19.5	6.0	R	62S	311E	28.9	9.1
Paganzo Village	La Colina		300°C	35(136)	221	+70	56.5	21.8	174	+56	11.0	3.3	R	81S	327E	13.4	4.0[e] F
Los Colorados, lower site	La Colina	Fine grained red sandstones	NRM	4(12)	57	+57	31.4	38.0	51	+69	31.3	38.0	R	13S	321E	43.5	56.8
Los Colorados, lower site	La Colina		500°C	4(12)	140	+60	10.9	12.4	161	+61	10.9	12.4	R	69S	335E	14.4	16.5
Los Colorados, middle beds[c]	La Colina		470 and 500°C	27(81)	152	+60	9.2	3.2	172	+63	9.1	3.1	R	74S	313E	13.0	4.5[e] E
Los Colorados, highest sites[d]	La Colina	Fine grained red sandstones and purple silts	NRM	13(39)	131	+58	15.5	8.1	180	+61	13.9	7.2	R	69S	335E	10.1	19.3
Los Colorados, highest sites	La Colina		350°C	13(39)	147	+54	7.5	3.9	171	+53	6.6	3.4	R	82S	349E	7.4	3.8[e] G
Upper Paganzo																	
Paganzo Village	Amana	Laminated sandstones and siltstones	NRM	22(88)	333	+52	76.1	54.2	130	+64	70.7	43.6	R	46S	340E	71.8	45.4
Paganzo Village	Amana		Thermally cleaned	7(36)	262	+67	43.3	36.8	171	+48	16.9	12.8	R	83S	15E	18.7	14.3[e] H
Rio Chasquil	Amana	Fine-medium grained sandstones	NRM	30(87)	221	+18	32.6	11.3	180	+60	35.8	12.7	M	73S	295E	43.3	16.0
Rio Chasquil	Amana		350°C	21(59)	205	+12	21.2	8.6	176	+40	18.8	7.5	M	85S	64E	18.9	7.6[e] I

[a] From Thompson and Mitchell [4].
[b] Includes data from Embleton [2].
[c] Includes data from Embleton [3].
[d] Possibly Upper Paganzo.
[e] Palaeopoles plotted in fig. 7.

N = number of samples, n = number of specimens. csd = circular standard deviation. R = directions of magnetization reversed with respect to earth's present magnetic field. M = directions of magnetization both normal and reverse.

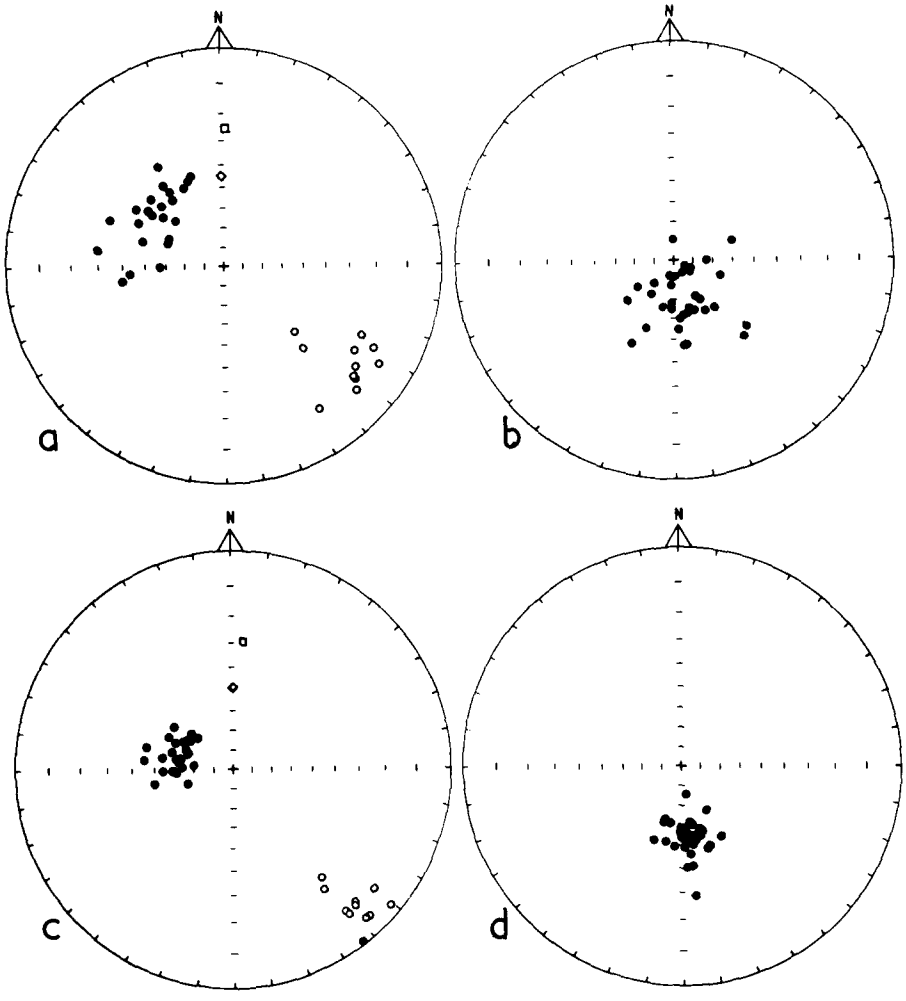


Fig. 3. Paganzo village. Sample mean directions; (a) and (b), natural remanent magnetization; (c) and (d) after thermal demagnetization; (a) and (c) referred to the present horizontal; (b) and (d) referred to the palaeo-horizontal; open square, present magnetic field direction; open diamond, present axial dipole field direction.

Thermal cleaning at 450°C gave the lowest scatter of directions and this is taken to give the best estimate of the primary directions of magnetization from the locality (table 2 and fig. 2). The directions of magnetization are significantly different to both the present day axial dipole and actual magnetic field directions. The fold test [9] cannot be applied effectively to give the relative ages of the magnetization as the beds from which the samples were collected lie at similar orientations. However, the magnetic directions, when referred to the palaeohorizontal, lie closer to the present magnetic field directions than when computed in their field orientations and with progressive cleaning the magnetic vectors move away from the present magnetic field directions. All the above results suggest that the magnetization which has been isolated at 450°C was not produced in the earth's present magnetic field. This magnetization gives a south palaeomagnetic pole position at 85°S, 335°E (A_{95} = 6.6°).

6. Paganzo Village

Seventyone hand samples from the La Colina and

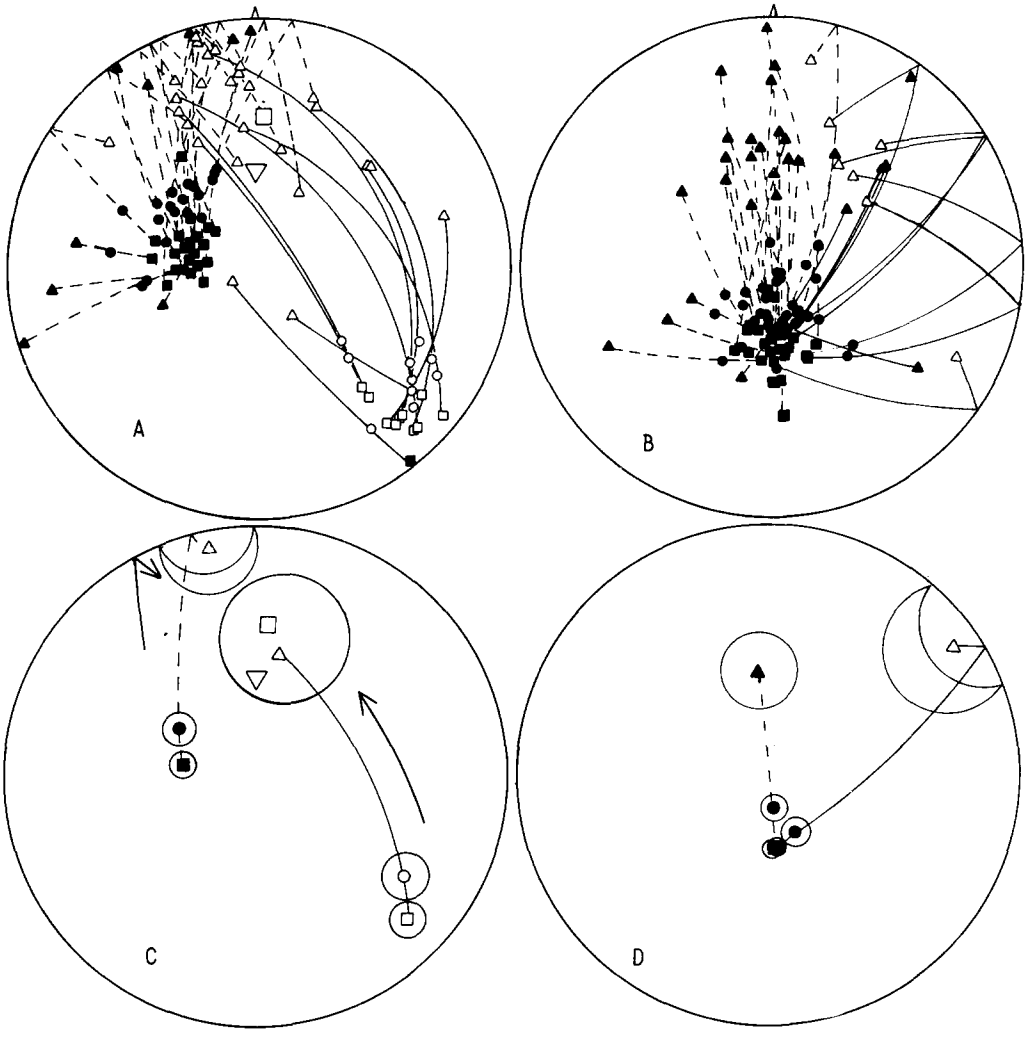


Fig. 4. Secondary directions of magnetization. Great circles connect secondary directions (triangles) through NRM directions (circles) to primary directions (squares). A and B, separate samples. C and D, sample means from each limb of anticline and circles of confidence. A and C referred to present horizontal. B and D referred to palaeo-horizontal. Large square and inverted triangle refer to present field and axial dipole field respectively.

Amana formations, and a basalt from near the base of the La Colina formation were collected from a well exposed sequence. The La Colina and Amana sediments were sampled from both limbs of an anticline. The age of the folding is possibly Triassic but most probably Tertiary and the anticline has a horizontal hinge axis (Azcuy and Morelli, personal communication).

The basalt collected from only one limb of the anticline has been previously studied by Embleton [10] and his results have been combined with new results from two sites near Las Torres (fig. 1). A radiometric age (295 my), has been reported previously [4] for the basalt which therefore lies close to the lower boundary of the Kiaman Magnetic Interval. Results from the basalt are summarized in table 2.

Whenever possible, four discs were cut from each of the thirtyfive samples of La Colina sandstones. One specimen from each sample was demagnetized in ten steps up to a temperature of 670°C. The remaining specimens were then demagnetized at 300°C.

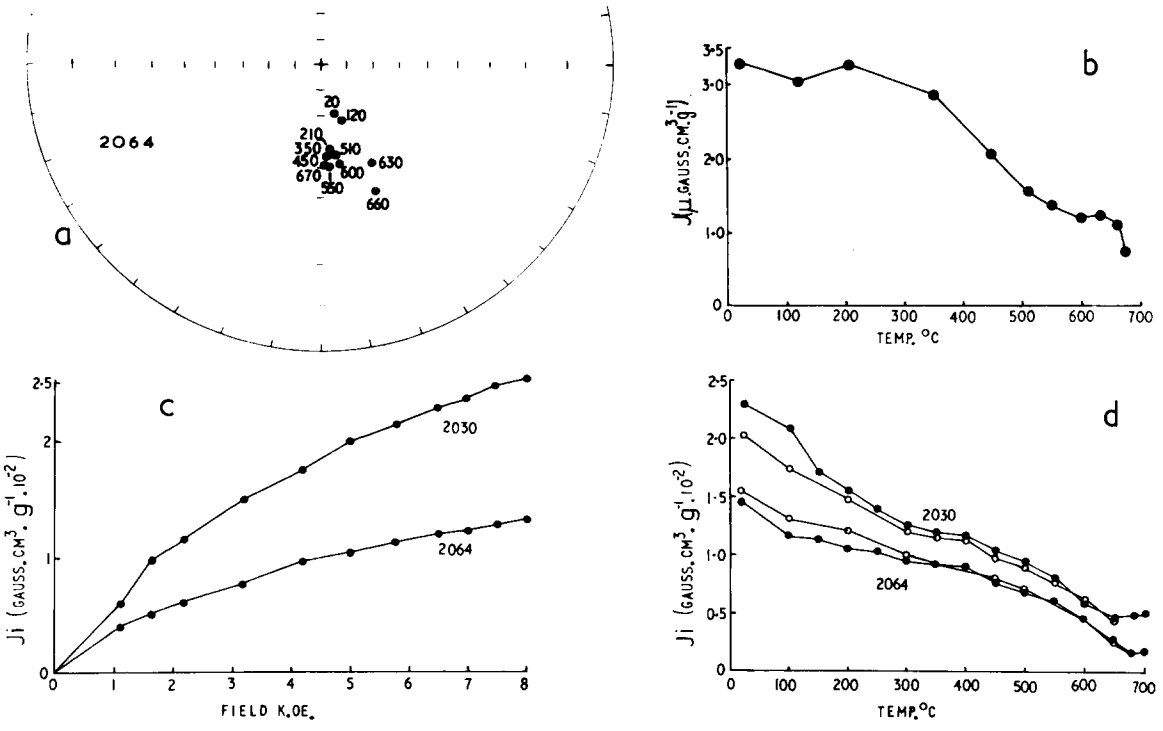


Fig. 5. Pilot specimens, La Colina formation, Paganzo village. (a) Change of direction of magnetization of pilot specimen with step thermal demagnetization. (b) Thermal decay curve of remanence of pilot specimens. The geometric mean intensity at each successive stage of demagnetization is shown. (c) J_i–H curves for two specimens, corrected for diamagnetic components. (d) J_i–T curves corrected for diamagnetic components. Measurements made at 7 KOe; full and open circles represent heating and cooling curves respectively.

All specimens were reversely magnetized and all lie stratigraphically above the basalt collected from the same locality. The specimens contain two components of magnetization, and the fold test can be used to double effect in determining their ages in relation to the formation of the anticline, viz. (1) to show the most stable direction was acquired prior to the folding, (2) the less stable magnetization was acquired after folding. The results are shown in figs. 3 and 4 and table 2. Before correction for bedding the directions of magnetization fall into two separate groups (figs. 3a and c) corresponding to samples collected from opposite limbs of the anticline. When the correction for the tilt of the beds is applied (figs. 3b and d), the directions from a single cluster. The most stable direction of magnetization was therefore locked in the sediments before folding took place. There is a suggestion (fig. 3a) that the NRM directions streak towards the present field directions, so it was decided to test whether the secondary magnetization was produced in the rocks before or after folding.

To obtain the secondary direction of magnetization in each sample the NRM was regarded as the vector sum of the primary plus the secondary directions, the magnetization remaining after demagnetization at 300°C being considered representative of the primary direction. The vector difference between the NRM and demagnetized directions should therefore yield the secondary directions. The primary (squares) secondary (triangles) and primary plus secondary (circles) directions and the great circles connecting them are plotted for each hand sample before and after a bedding correction has been applied (figs. 4a and b). The mean directions and their circles of 95% confidence for samples from opposite limbs of the anticline are plotted in figs. 4c and d.

The secondary directions fall in two groups when referred to the palaeohorizontal (figs. 4b and d) but

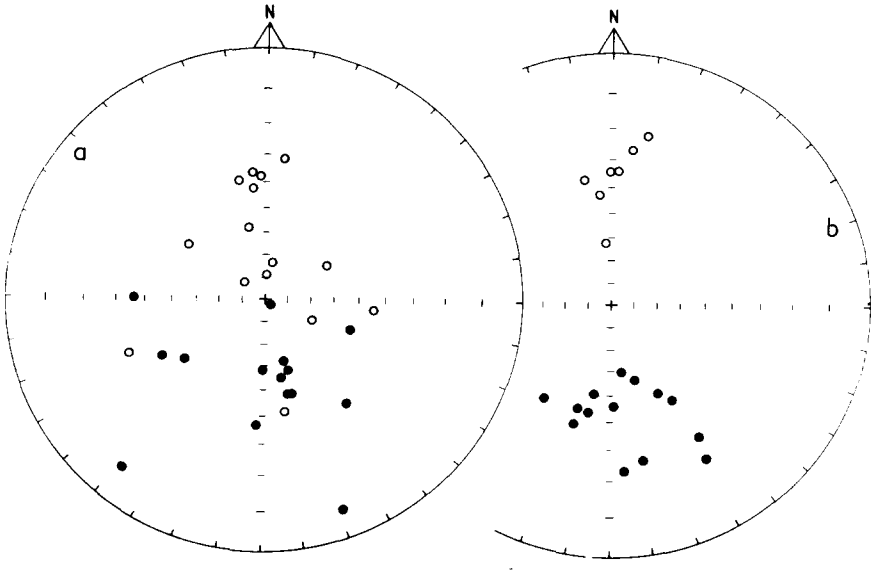


Fig. 6. Rio Chasquil sample mean directions. Key as for fig. 2.

lie nearer together, close to the present earth's magnetic field directions before correction for bedding (figs. 4a and c). An additional point is that although demagnetization at 300°C yields the best estimate of the primary direction, i.e. when all the secondary component has been removed, it is not the best estimate of the magnitude of the primary direction, because part of the primary magnetic vector has been removed during thermal cleaning of the secondary magnetization. An example of this is shown in fig. 5a where the secondary direction has been eliminated above 120°C, and further demagnetization to 300°C simply removes part of the primary magnetization. This means that the true directions of secondary magnetization lie slightly farther along the great circles, in the direction of the arrows, in fig. 4c and would then give an even better single grouping of directions. (The great circles in fig. 4b are streaking towards a normal primary direction of magnetization, but this trend is solely a reflection of the geometrical property of great circles which must pass through the antipodal point).

It is concluded that the secondary direction of magnetization was produced in the rocks after the formation of the anticline and probably in the earth's present magnetic field directions, and that the primary direction was acquired before folding. This primary magnetization gives a south palaeomagnetic pole at 81°S 33°W, $A_{95} = 4.0°$.

J_i vs. field and J_i vs. temperature measurements were made to help determine the magnetic component in the sediments. Results for two typical samples are shown in figs. 5c and d. The gradient of the J_i–H curves in fig. 5c shows that the magnetic material is not saturated at 8000 Oe and indicates the presence of haematite. Curie temperatures close to that of haematite (680°C) are shown in fig. 5d. Both methods suggest the carrier of the magnetic remanence is haematite in agreement with the demagnetization results of the pilot specimens (fig. 6b) which have a blocking temperature equal to a haematite Curie temperature.

Of the twentytwo hand samples investigated from the Amana formation only seven showed high stability when thermally cleaned. These seven cleaned results give a south palaeomagnetic pole at 83°S 15°C, A_{95} = 14°. Again only reverse directions are recorded and the fold test indicates stability of the remanence.

7. Los Colorados

The collection from Middle Paganzo sediments at this locality, started by Embleton in 1967 [3], was extended by twentyfive samples from a higher stratigraphical level. The lowest section of the author's succession followed on immediately from the highest level of Embleton's collection. The author's lowest beds

are of Middle Paganzo age (La Colina formation), but the highest beds may reach into the Amana formation (Upper Paganzo age).

As expected, the magnetic properties of the samples from the author's oldest beds were very similar to those described for Embleton's youngest beds. They exhibit similar directions (reversed) and intensities of remanence before and after thermal demagnetization and both collections contain a weak secondary magnetization which was acquired in the earth's present magnetic field. In all except seven samples this secondary magnetization was eliminated by thermal demagnetization; these seven samples were not used in the calculation of mean pole positions.

The results from the stratigraphically lowest site have been grouped with Embleton's "upper beds" to give a south palaeomagnetic pole at 74°S 47°W, A_{95} = 4.5° (table 2). The stratigraphically higher localities yield a south palaeomagnetic pole at 82°S 11°W, A_{95} = 3.8°.

8. Rio Chasquil

A well-exposed succession of uniformly dipping red sandstones was sampled at this locality in the north of the Paganzo Basin (fig. 1). Eightyseven specimens were cut from thirty hand samples. The lower part of the succession was found to be normally magnetized and is separated by a fault from an upper reversely magnetized sequence. The initial mean sample directions are shown in fig. 6a. The geometrical mean intensity is 3.4 μG. Eleven pilot specimens were demagnetized in steps of 100°C between 150°C and 650°C, showing a range of blocking temperatures above 400°C. Thermal demagnetization was completed on the remaining specimens at a temperature of 350°C. Twelve samples exhibited a considerably higher within-sample scatter (> 62°) than the other samples. In three of these samples the scatter was due to one aberrant direction which was rejected. In the other nine samples all specimen directions were scattered. It was concluded that thermal demagnetization had been unable to isolate the primary direction and the samples were excluded in the calculation of the mean.

The normal and reversed directions (fig. 6b) were combined to give an overall mean south palaeomagnetic pole position at 85°S 64°E, A_{95} = 7.6°. The presence of the normal and reverse directions suggests that the magnetization is of a post-Kiaman age, i.e. younger than about 235 my.

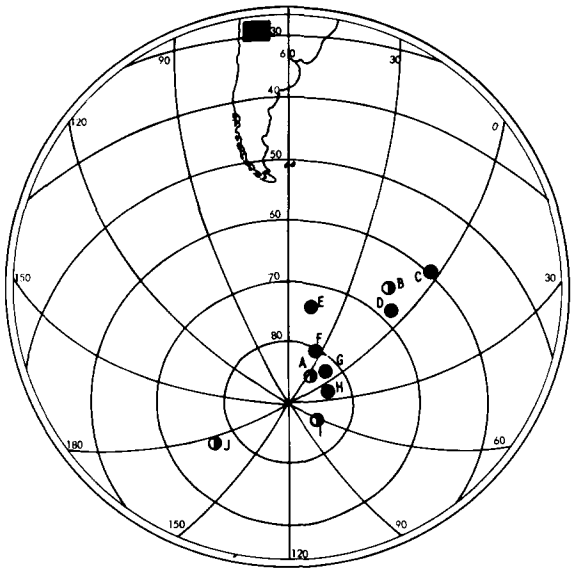


Fig. 7. South palaeomagnetic pole positions of cleaned samples from the Paganzo Basin. Full circles, reverse magnetization; half full circles, normal and reverse magnetization. A, Bum-Bum; B, La Colina basalt; E, Los Colorados middle beds; F, Paganzo village, La Colina formation; G, Los Colorados upper beds; H, Paganzo village, Amana formation; I, Rio Chasquil; all from table 2. D, Huaco gorge and J, Villa Union from Embleton, Ph. D. Thesis; C, Los Colorados lowest beds, from Embleton [2].

9. Discussion of results

The palaeomagnetic results from the three formations of the Paganzo succession have been shown to contain magnetically stable components by both field and laboratory tests. The K-Ar dating study associated with this work [4] combined with geological evidence and the direction and polarization of the magnetization closely limit the age of the remanence. The results are in good agreement with those obtained from previous work in South America [1, 2] and both are summarized in fig. 7. The lower and upper boundaries of the Kiaman Magnetic Interval have now been located in the Paganzo sequence. It is clear that rocks of the La Colina formation (Middle Paganzo age) hold a remanence of Kiaman age.

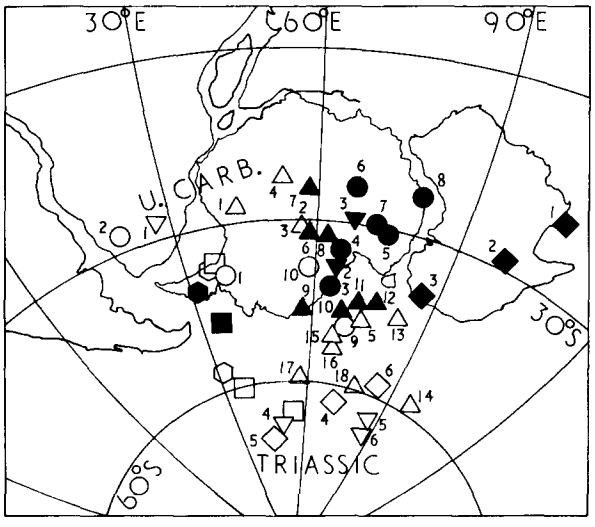


Fig. 8. Stereographic projection of Upper Carboniferous to Triassic south palaeopoles plotted on Smith and Hallam's reconstruction of Gondwanaland. Gondwana poles listed in table 3. Laurasian poles are located using Bullard et al.'s [23] reconstruction. Shaded symbols represent poles of Kiaman (reversed) magnetization. South American, Australian, African and Indian poles referred to by upright triangles, circles, inverted triangles and diamonds respectively. Eurasian and North American mean poles by hexagons and squares respectively.

In reviewing palaeomagnetic results from the southern continents, some authors (e.g. see refs. [11, 12]) have attributed an age other than Kiaman to Middle Paganzo palaeomagnetic results and have only matched African and Australian Kiaman results. A sensible polar-wander path was obtained by using a reinterpretation of the African K3 redbeds [13]. However, Opdyke (personal communication) now favours his original interpretation of the African K3 redbeds (i.e. the magnetization of both the Songwe-Kiwira and Ketewake-Mchuchuma beds and the Galula beds are of Kiaman age), [14], and the author uses Opdyke's original palaeomagnetic pole (38°S 65°E) when discussing the African results.

Palaeomagnetic data can be used with effect to delimit the past reconstructions of continents in two ways, (1) to assemble the continental fragments using palaeomagnetic data alone (with the restriction that for a particular period only the latitude of any continent is fixed), and then to assess the plausibility of the reconstruction with geological evidence or (2) to use the palaeomagnetic evidence to test continental fits deduced independently of palaeomagnetic results (e.g. by topographic fits). The author considers the second approach is more profitable with the palaeomagnetic data available at present for the Carboniferous to Triassic periods. Palaeomagnetic poles from the various continents over the period of time spanned by Paganzo sedimentation were taken from the pole lists included in Creer's recent review of palaeomagnetism [15]. The Carboniferous to Triassic poles from his tables 4 to 7 which were used to calculate the "period means", combined with the pole positions obtained in this investigation (table 2) and the North American and Eurasian mean Upper Carboniferous to Triassic palaeomagnetic poles from Creer's tables 4 and 10 were plotted on several previously proposed fits of Pangea.

To illustrate the results, the reconstruction proposed by Smith and Hallam [16] is used as shown in fig. 8. Palaeomagnetic poles from Gondwana continents plotted in fig. 8 are listed in table 3. The Kiaman magnetic poles (shaded) fall in a cluster between the Upper Carboniferous and the Triassic poles. The same polar movement from north to south is evident for all the continents but the Laurasian poles lie distinctly to the west of the Gondwana groups. The cause of this separation of the poles has been discussed by several authors, e.g. due to an alternative fit along Tethys, Van Hilten [17]; by inclination error and/or a short period of continental drift, Creer [15]; and due to long-term non-axially central dipolar components of the earth's magnetic field, Briden et al. [18].

The general north–south movement of the pole is however similar for all the continental blocks. Whether this movement of about 0.3 cm/yr is polar wander or simply plate motion of Gondwanaland and Laurasia as a whole is not yet distinguishable for this period. A possible distinguishing factor would be that for plate motion one would expect the leading edge of the plate (Alaska?) to have been a more active orogenic region than the trailing edge (Western Antarctica?).

Considering the palaeomagnetic results from the southern continents, the type of reconstruction favoured by McElhinny [11, 19], in particular the separation of Australia and Antarctica, is difficult to reconcile with the South American results. The Australian pole positions fall to far to the north-east of the

Table 3
Summary of Gondwana Middle/Upper Carboniferous to Triassic palaeomagnetic data.

S. America			Australia			Africa			India		
Pole no.	Rock unit	Pole position	Pole no.	Rock unit	Pole position	Pole no.	Rock unit	Pole position	Pole no.	Rock unit	Pole position
Pre-Kiaman											
1	Taiguati	54S 26W	1	Paterson Toscanite	73S 147E	1	Dwyka varves	27S 27E			
2	Piaui	65S 13W	2	Lr. Kuttung Tosc.	84S 46E						
3	Red beds Arg.	65S 13W									
4	Piaui	55S 9W									
5(A)	Bum-Bum	85S 25W									
Kiaman											
6(B)	La Colina basalt	66S 12W	3	Main Glacial	53S 149E	2	K3 redbeds	38S 65E	1	Talchir	32N 134E
7(C)	Los C. lowest	60S 3W	4	Rocky Creek	51S 138E	3	Nuanetsi Lavas	29S 68E	2	Kamthi beds	21N 130E
8(D)	Huaco gorge	68S 6W	5	Currabubulla	42S 135E				3	Mangli beds	7N 124E
9(E)	Los C. middle	74S 47W	6	L. Marine basalt	46S 122E						
10(F)	Paganzo (La Colina)	81S 33W	7	U. Marine basalt	44S 132E						
11(G)	Los C. Upper	82S 11W	8	Moonbi	35S 128E						
12(H)	Paganzo (Amana)	83S 15E									
Post-Kiaman											
13(I)	Rio Chasquil	85S 64E	9	Chocolate Shale	49S 160E	4	Red sst Zambia	68S 50E	4	Kamthi sst	18S 127E
14(J)	Villa Union	77S 175E	10	Brisbane Tuff	57S 143E	5	Shawa Ijolite	64S 86E	5	Parsora	30S 125E
15	Motuca	81S 64W				6	U. Beaufort	67S 87E	6	Pachmarhi	10S 130E
16	S. Nihuil	81S 78W									
17	Uspallata	74S 94W									
18	Los Tenos	80S 132W									

Pole numbers referred to in fig. 8, letters in fig. 7.

South American and African results. This difficulty is reduced when the reconstruction proposed by Smith and Hallam [16] is used as shown in fig. 8. However, the Australian and Indian poles still lie to the northeast of the South American pole positions.

Flores [20] has recently suggested on geological grounds that Madagascar would be better placed to the south of its present position. Green (personal communication) has shown that magnetic lineations between Africa and Madagascar lend strong support to the interpretation that Madagascar has moved first eastwards and then northwards from a southern position [21]. However Heirtzler and Burroughs [22] have concluded from bathymetric work in the Mozambique Channel that Madagascar has moved from the north, but there is no confirmation of this from magnetic lineations. If Madagascar has moved from the south, this probably means that Australia, India and Antarctica should be swung clockwise to a position further south than illustrated in fig. 8 and would explain the apparent discrepancy between the South American and Australian results.

Acknowledgements

I wish to thank Professor K.M. Creer for supervising the work, Dr. B.J.J. Embleton for help and advice with fieldwork, Dr. D.H. Tarling for critically reading the final manuscript and N.E.R.C. for financial support.

References

[1] K.M. Creer, A Palaeomagnetic survey of South American rock formations, Phil. Trans. Roy. Soc. London. A267 (1970) 457.
[2] B.J.J. Embleton, Palaeomagnetic results for the Permian of South America and a comparison with the African and Australian Data, Geophys. J. 21 (1970) 105.
[3] B.J.J. Embleton, Ph. D. thesis (University of Newcastle upon Tyne) 1970.
[4] R. Thompson and J.G. Mitchell, Palaeomagnetic and radiometric evidence for the age of the lower boundary of the Kiaman Magnetic Interval in South America, Geophys. J. (in press).
[5] C.L. Azcuy and J.R. Morelli, Geologia de la Comarca Paganzo-Amana el grupo Paganzo formaciones que lo Componen y sas relaciones, Rev. Assoc. Geol. Arg. XXV (1970) 405.
[6] D.W. Collinson, The astatic magnetometers at Newcastle upon Tyne, in: Methods of Palaeomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds. (Elsevier, Amsterdam, 1967) 60.
[7] D.W. Collinson and L. Molyneux, An instrument for the measurement of isotropic initial susceptibility of rock samples, in: Methods in Palaeomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds. (Elsevier, Amsterdam, 1967) 368.
[8] D.W. Collinson and A. de Sa, A new magnetometer for small scale magnetic studies, J. Phys. E. (1971) 337.
[9] J.W. Graham, The stability and significance of magnetism in sedimentary rocks, J. Geophys. Res. 54 (1949) 131.
[10] B.J.J. Embleton, Resultados palaeomagneticos del basalto de la base de la formacion La Colina (Paganzo Medio), La Rioja, Republica Argentina, Rev. Assoc. Geol. Arg. XXV, 3 (1970) 347.
[11] M.W. McElhinny, Formation of the Indian Ocean, Nature 228 (1970) 977.
[12] M.W. McElhinny and J.C. Briden, Continental drift during the Palaeozoic, Earth Planet. Sci. Letters 10 (1971) 407.
[13] M.W. McElhinny and N.D. Opdyke, Palaeomagnetism of some carboniferous glacial varves from Central Africa, J. Geophys. Res. 73 (1968) 689.
[14] N.D. Opdyke, The palaeomagnetism of the Permian redbeds of southwest Tanganyika, J. Geophys. Res. 69 (1964) 2477.
[15] K.M. Creer, A review of palaeomagnetism, Earth Sci. Rev. 6 (1971) 369.
[16] A.G. Smith and A. Hallam, The fit of the Southern Continents, Nature 225 (1970) 139.
[17] D. Van Hilten, Evaluation of some tectonic hypotheses by palaeomagnetism, Tectonophysics 1 (1964) 1.
[18] J.C. Briden, A.G. Smith and J.T. Sallomy, The geomagnetic field in Permo-Triassic time, Geophys. J. 23 (1970) 101.
[19] M.W. McElhinny and G.R. Luck, Palaeomagnetism and Gondwanaland, Science 168 (1970) 830.
[20] G. Flores, Suggested origin of the Mozambique Channel, Trans. Geol. Soc. Africa 73 (1970) 1.
[21] A. Green, Sea floor spreading in the Mozambique Channel (in press).
[22] J.R. Heirtzler and R.H. Burroughs, Madagascar's palaeoposition; new data from the Mozambique Channel, Science 174 (1971) 488.
[23] E.C. Bullard, J.E. Everett and A.G. Smith, The fit of continents around the Atlantic, Phil. Trans. R. Soc. A, 258 (1965) 41.